

Input Content (IC)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Empirical dataset inspection partially resolves what the paper does not document. Ethiopian calendar months (Meskerem, Hamle) appear as DATE slot values [DATASET-D1, D2, D6], Ethiopian intercity cities (Dire Dawa, Bahir Dar, Hawassa, Arba Minch) appear as CITY_OR_PROVINCE slots [DATASET-D2, D9], and Ethiopia-specific financial entities (Amhara Bank, Commercial Bank of Ethiopia, TeleBirr, Dashen Bank) appear in banking-domain instances [DATASET-D3, D4, D5, D34]. This is materially better than the registry feared. However, several deployment-critical content gaps remain: (a) Ethiopian Birr (ETB) was absent as a CURRENCY/MONEY slot value in any sampled Amharic example despite Naira/Rand appearing in other tracks [DATASET-D5, D19, D22 vs Amharic absence]; (b) most Amharic DATE slots used relative expressions rather than named Ge'ez months (only 2/9) [DATASET-D15, D16, D28]; (c) Ethiopian time convention (6-hour offset) is unaddressed in TIME annotations [DATASET-D11, D23]; (d) no code-switched Amharic–Oromo / Amharic–Tigrinya utterances appear, despite ~70% L2 Amharic users in the deployment population [DATASET-D1–D18 monolingual]. IC was marked HIGH priority in elicitation, so these residual gaps weigh heavily.

ID	Evidence
D1	ከድረዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D2	target: "CITY_OR_PROVINCE: ድረዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም እምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D3	የአማራ ባንክ አካውንቱ ስንት አለው? / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated
D4	ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ? / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank
D5	በቴሌብር የብድር አካውንቱ ላይ ... / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform
D6	ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ? / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar
D7	ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንገረኝ? / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D9	raw: "ለቀጣይ የጉብኝት ጉዞ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding
D10	raw: "ከቴሌብር አካውንቱ ገንዘብ ለማን ለማን እንደላኩ ዝርዝሩን አሳየኝ?" / text: "Show me history of my telebirr transfers?" — TeleBirr payment company referenced in Amharic utterance
D11	ነገ ጠዋት 12 ሰአት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰአት"
D12	Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyya irraa naaf kuti — "Book me an Ethiopian Airlines flight from Dire Dawa to Bole then to Jimma." — spans empty despite city names and airline
D13	Ebila 12 hanga Ebila 14tti Mattuu keessatti kireeffannaa konkolaataa argachuun ni danda'amaa / target: "DATE: Ebila 12 hanga Ebila 14 \$\$\$ CITY_OR_PROVINCE: Mattuu — Oromo month name "Ebila" (April) used as date slot — culturally grounded Ethiopian Oromo calendar

D14	Sagantaa sirna cidhaaf sadaasa 21tti qabame kalaandarii koo irraa haqi maloo" / target: "CALENDAR_EVENT: sirna cidhaa \$\$ DATE: sadaasa 21 — Oromo month "Sadaasa" (November in Oromo calendar) used as date slot
D15	የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ::" / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name
D16	ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መከራየት እፈልጋለሁ::" / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month
D17	በቶቶት ለቃለአብ ቦታ ማስያዝ እንዳለብኝ..." / target: "RESTAURANT_NAME: ቶቶት \$\$ PERSONAL_NAME: ቃለአብ — Restaurant "ቶቶት" (Totot) — Ethiopian cultural reference; personal name "ቃለአብ" is distinctly Ethiopian
D18	የቦሎ ማሳደሻ ስንት ነው" / target: "BILL_TYPE: የቦሎ ማሳደሻ — "What is the Bole maintenance/upgrade fee?" — "ቦሎ" likely refers to Bole (Addis Ababa district)
D19	Naira nawa gare ni ... na bankin Gt?" / target: "CURRENCY: Naira \$\$ BANK_NAME: Gt — Nigerian Naira currency annotated in Hausa — confirms per-language currency grounding
D22	Ndinganayo imali engange R700 kwiakhawunti yam yaseStandard Bank?" / target: "MONEY: R700 \$\$ BANK_NAME: yaseStandard Bank — South African Rand (R700) and Standard Bank in Xhosa
D23	"ነገ ጠዋት 12 ሰአት..." (no span annotation) — "12 ሰአት" (12 o'clock) unannotated — "ጠዋት" (morning) absent from target; Ethiopian time convention unclear
D28	በዚህ ሳምንት ምንም ቀጠሮ እንዲኖረኝ አልፈልግም::" / target: "DATE: በዚህ ሳምንት — Relative week reference, no Ethiopian calendar month
D34	raw: "እባካችሁ የዳሽን ባንክ ሂሳቤን ታረጋግጡልኛለችሁ?" / text: "can you please check my dasion bank account" — "Dashen Bank" (major Ethiopian bank) — Amharic source is culturally grounded but English transliteration is lossy ("dasion")

Strengths

- Ethiopian calendar months (Meskerem, Hamle) confirmed as valid DATE slot annotations in travel-domain Amharic instances [DATASET-D1, D2, D6]
- Major Ethiopian intercity cities (Dire Dawa, Bahir Dar, Hawassa, Arba Minch) appear as CITY_OR_PROVINCE slots [DATASET-D2, D9]
- Ground-up cultural elicitation methodology rather than translation [Q6, Q13, Q30]
- Ethiopia-specific banks and TeleBirr appear as natively grounded entities [DATASET-D3, D4, D5]

ID	Evidence
Q6	"The data construction process starts with providing an annotator with sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, and they are to come up with culturally-relevant similar sentences and relevant slot entities (see Figure 1)." (p.1)
Q13	"However, these efforts face two key challenges: (1) the translationese effect, which makes utterances sound less natural in the target languages (Vanmassenhove et al., 2021; Bizzoni et al., 2020), and (2) the creation of utterances that are less culturally relevant." (p.1)
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ:: — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$ CITY_OR_PROVINCE: ባህርዳር \$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D3	የአማራ ባንክ አካውንቱ ስንት አለው" / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated
D4	ንግድ ባንክ ላይ የቀረኝ ንገዛብ አለ" / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank
D5	በቴሌብር የብድር አካውንቱ ላይ ..." / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform
D6	ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar
D9	raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — deployment demands Amharic city names/abbreviations, Ethiopian calendar dates, Birr fares, and operator-specific terminology, confirmed by elicitation Q2 (HIGH priority) and YAML calendar/toponym sections.

IC-2: Partially. Cultural grounding is confirmed for banking-domain Amharic [DATASET-D3, D4, D5] and partially for travel (Ethiopian cities + Meskerem present) [DATASET-D1, D2, D9]. The grounding methodology is documented [Q30, Q40, Q41].

IC-3: Western-content leakage was not observed in the Amharic sample (though present in some other tracks, e.g., Hausa Taco Bell [DATASET-D31]). Air-travel-to-international-destinations framing dominates the Amharic travel domain [DATASET-D7, D8].

IC-4: Native Amharic speakers annotated [Q7], but regional/L2 demographics undisclosed and code-switched utterances absent from sample [DATASET-D1–D18]; full re-annotation by Ethiopian regional pool would be advisable.

IC-5: Documented residual content gaps: (a) Birr/ETB absent from sampled Amharic CURRENCY/MONEY slots; (b) Ethiopian calendar months sparse vs. relative dates [DATASET-D15, D16, D28]; (c) Ethiopian clock convention undocumented; (d) no code-switched utterances [DATASET-D1–D18].

ID	Evidence
Q2	"Intent detection and slot-filling are crucial components of the natural language understanding module in task-oriented dialogue systems (Hemphill et al., 1990; Coucke et al., 2018; Gupta et al., 2018)." (p.1)
Q6	"The data construction process starts with providing an annotator with sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, and they are to come up with culturally-relevant similar sentences and relevant slot entities (see Figure 1)." (p.1)
Q7	"utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)
Q40	"A Xhosa annotator was asked to generate another utterance belonging to the same intent type but capturing the South African context where the language is spoken." (p.3)
Q41	"Thus, the annotator used the R200 as "money" with currency Rand (R), and more familiar South African banks such as "FNB" and "Absa" for "bank name" slot." (p.3)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$ CITY_OR_PROVINCE: ባህርዳር \$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D3	የአማራ ባንክ አካውንቱ ስንት አለው" / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated
D4	ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ" / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank
D5	በቴሌብር የብድር አካውንቱ ላይ ..." / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform
D6	ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar

D7	ወደ ቱርክ በፍጥነት የምደርስበት የበረራ አማራጭ ንገረኝ። / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D9	raw: "ለቀጣይ የጉብኝት ጉዞ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding
D10	raw: "ከቴሌብር ብር አካባቢ ገንዘብ ለማን እንደላኩ ዝርዝርን አሳየኝ?" / text: "Show me history of my telebirr transfers?" — TeleBirr payment company referenced in Amharic utterance
D11	ነገ ጠዋት 12 ሰአት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰአት"
D12	Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyaa irraa naaf kuti — "Book me an Ethiopian Airlines flight from Dire Dawa to Bole then to Jimma." — spans empty despite city names and airline
D13	Ebila 12 hanga Ebila 14tti Mattuu keessatti kireeffannaa konkolaataa argachuun ni danda'amaa / target: "DATE: Ebila 12 hanga Ebila 14 \$\$\$ CITY_OR_PROVINCE: Mattuu — Oromo month name "Ebila" (April) used as date slot — culturally grounded Ethiopian Oromo calendar
D14	Sagantaa sirna cidhaaf sadaasa 21tti qabame kalaandarii koo irraa haqi maloo / target: "CALENDAR_EVENT: sirna cidhaa \$\$\$ DATE: sadaasa 21 — Oromo month "Sadaasa" (November in Oromo calendar) used as date slot
D15	የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ። / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name
D16	ከሚመጣው ወር ጀምሮ ለ3 ወር ትየታ መኪና መከራየት እፈልጋለሁ። / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month
D17	በቶቶ ለቃለኡብ ቦታ ማስያዝ እንዳለብኝ..." / target: "RESTAURANT_NAME: ቶቶ \$\$\$ PERSONAL_NAME: ቃለኡብ — Restaurant "ቶቶ" (Totot) — Ethiopian cultural reference; personal name "ቃለኡብ" is distinctly Ethiopian
D18	የቦሎ ማሳደሻ ስንት ነው / target: "BILL_TYPE: የቦሎ ማሳደሻ — "What is the Bole maintenance/upgrade fee?" — "ቦሎ" likely refers to Bole (Addis Ababa district)
D28	በዚህ ሳምንት ምንም ቀጠሮ እንዲኖረኝ አልፈልግም። / target: "DATE: በዚህ ሳምንት — Relative week reference, no Ethiopian calendar month
D31	Za ku iya tabbatar min ina da ajiyar tebur a taco bell da karfe 7 na yamma / target: "RESTAURANT_NAME: taco bell — "Taco Bell" — US fast-food chain, Western entity in Hausa utterance

Information Gaps

- Birr coverage in the 99.4% of Amharic data not sampled
- Frequency of named Ethiopian calendar months vs. relative dates in full travel split
- Whether any code-switched Amharic utterances exist in unsampled portions

Requires Expert Verification

- Native Amharic-speaking expert review of Ethiopian time convention handling in TIME slots across the full dataset

Remediation

Gap 1: Ethiopian Birr (ETB) absent as CURRENCY/MONEY slot in sampled Amharic; most DATE slots use relative expressions rather than named Ge'ez months; Ethiopian clock convention undocumented; no code-switched utterances

Recommendation: Construct a 'ETH-Travel-Slot Stress Set' of Amharic utterances densely populated with (a) Birr fare amounts (VIP/economy tiers), (b) Ge'ez calendar month names across all 13 months including Pagume, (c) Ethiopian-clock TIME values with explicit annotator convention, (d) code-switched Amharic–Oromo and Amharic–Tigrinya tokens for cities and booking terms; evaluate alongside INJONGO Amharic test.

Gap 2: Annotation gaps observed empirically (empty spans on temporally explicit utterances)

Recommendation: Run a targeted audit pass over Amharic train/dev/test for empty-span utterances containing temporal lexemes ('ሰአት', 'ጠዋት', 'ነገ', Ge'ez month names) and patch with corrected annotations before using F1 as a model-selection metric.